

“What is our collective vision of a desirable and achievable educational system that leverages automation to advance learning while protecting and centering human agency?”
(United States Department of Education, 2023)

Artificial Intelligence in Education

I. Guiding Principles for AI Use in Education

The use of Artificial Intelligence (AI) is quickly and powerfully transforming education, but that speed and power comes with great responsibility. As such, guidelines should distinguish between AI as a learning aid versus a replacement for thinking. The following list provides some guiding principles for responsible AI use in education to address this. These guiding principles are based on the U.S. Department of Education’s (ED) Office of Educational Technology report, [Artificial Intelligence and the Future of Teaching and Learning: Insights and Recommendations](#) (2023), and the National Center on Education and the Economy’s [Framework for AI-Powered Learning Environments](#) (2024).

Vision and Values	Having an agreed upon vision and set of values for education helps an organization prepare for future technological innovations and disruptions. The use of AI should serve the vision and values of the organization.
Center People	People, not machines, should be at the center of decision-making, and educators, students and families should retain their agency as the primary decision-makers. Human interactions and expertise will always be an essential part of learning. New technological innovations and disruptions make the role of teachers even more important, especially in relation to guiding students’ development of critical thinking, collaboration, and application of knowledge.
Advance Equity	Technology innovations should advance equitable access and opportunity. This is exemplified in the concept of digital equity, defined by the National Digital Inclusion Alliance as “a condition in which all individuals and communities have the information technology capacity needed for full participation in our society, democracy and economy.” Equity in AI involves ensuring fair access to technology and acknowledging the potential presence of biased data within AI systems.
Ensure Safety, Ethics, and Effectiveness	Data privacy, security and content appropriateness should be primary considerations when adopting new technology. Consider existing policies related to these issues and how the use of AI fits into existing frameworks.
Continuous Improvement	Decision-makers need to understand how AI models work so they can anticipate limitations, problems, and risks. Leaders should create a culture of continual evaluation and innovation to be ready to respond to future technological innovations and disruptions.

II. Learning with AI

AI provides significant opportunities and serious challenges for education. ED’s Office of Educational Technology report offers a guiding question: “What is our collective vision of a desirable and achievable educational system that leverages automation to advance learning while protecting and centering human agency?” (2023). It is important for teachers to consider this question, along with the questions listed below, when selecting an AI tool or website to use with students:

- Is the application age appropriate?
- What value does the tool bring to student learning?
- What are the unintended consequences and/or the impacts on the learning environment?
- How is data collected and used, and does it meet data practices standard?
- Before purchasing AI tools, have the specific needs and framework been identified to guide how technology will be used?

The table below highlights some ways AI can be used and some challenges to consider with implementation in learning environments. This is not an exhaustive list, and the specific opportunities and challenges will vary depending on the context and goals of each school and district. Please note that some of these opportunities and challenges may have more impact than others.

Opportunities	Challenges
Personalized Learning: AI can tailor learning experiences to individual student needs, pace, strengths, culture, and interests, promoting deeper understanding and engagement.	Bias and Discrimination: AI technologies, including those used in education, can perpetuate, and even amplify existing biases if not carefully designed and implemented. Thoughtfully crafting AI prompts can help improve results.
Improve Accessibility: AI can translate lessons and provide language supports for English Learners. It can also generate audio descriptions and offer other accessibility supports for students with disabilities.	Access and Equity: Access to AI tools may vary across families and schools, potentially widening the digital divide and creating new inequities.
Adaptive Instruction: AI-powered formative assessments can identify student learning gaps and adjust instruction in real-time, offering targeted support and enrichment. Human educators can enhance their impact on student learning by using valid AI tutoring tools to tailor instruction to individual student-needs.	Role of Teachers: Concerns exist about the evolving roles of teachers and the technology skills they’ll need to acquire, highlighting the need for upskilling and redefining the role of human interaction in AI-enhanced classrooms.
Automated Tasks: AI can support administrative tasks like grading, scheduling and data analysis, freeing up teacher time for more personalized interactions.	Teacher Training and Support: Educators need training and support to effectively integrate AI tools into their practices, maximize their benefits, and remove bias.

Opportunities	Challenges
Immersive Learning: AI can create virtual reality experiences and simulations, bringing abstract concepts to life and enhancing engagement.	Overreliance on Technology: Excessive dependence on AI could reduce critical thinking, creativity and social interaction in the classroom.
Early Intervention: AI can analyze student data to identify learning difficulties early on, enabling swift intervention and support.	Data Privacy and Security: Ethical concerns surround student data collection, storage, and use, requiring robust privacy policies and safeguards. (See Minn. Stat.13.32 [2026] Educational Data)
Career Exploration: AI can help students learn about future careers, assess their skills and connect them with relevant opportunities.	Limited Transparency: Understanding how AI systems work and make decisions can be vague and not well understood, raising concerns about accountability and control.
Creative Innovation: AI can be used to create new applications, programs, creative works and other tools. Individuals can be equipped to design and innovate with AI.	Lack of Standardized Regulations: Clear guidelines and regulations are needed to ensure ethical and responsible development and use of AI in education.

III. Creating District-level Guidance

Begin with What Already Exists

As with any emerging technology, AI is changing rapidly, and the answers won't be available right away. Begin with existing guidance, procedures, and policies to address immediate challenges. Consider the impact of the use of AI, particularly in:

- Data privacy
- Assessment
- Academic integrity

Work with all invested groups to develop further guidance for the use of AI anchored in the mission, vision, and values of the organization.

Prioritize Staff Learning

Centering people is a guiding principle for AI in education. People should be at the center of decision-making in how and when AI is used. To do this, schools should provide differentiated professional learning opportunities for not only teachers, but all staff.

School leaders should address risks and develop guidelines for the use of AI to support student learning. Without clear boundaries, students may rely on AI to complete tasks they should be practicing themselves, by passing on the

opportunities and repetition necessary to build foundational skills and cognitive capacity. Guidelines should distinguish between AI as a learning aid versus a replacement for thinking, ensuring students develop genuine understanding rather than simply producing AI-assisted work. Schools must address issues of equity, academic integrity, and assessing learning by having conversations with all members of the community before establishing clear expectations that prepare students to use AI ethically and effectively as adults.

Encourage Innovation and Knowledge Sharing

To encourage thoughtful innovation and collaborative exploration of AI use in schools, consider the following ideas:

- Include student voice in the creation of the district/school AI guidance.
- As educators, staff, and students explore and develop new ways of using AI applications, create opportunities for these ideas to be shared, refined, and built upon by others.
- Promote AI literacy for all learners, including how AI works and how it affects the world around us.
- Encourage open discussion about the benefits and risks of AI; outright bans can shut down opportunities for learning and growth.
- When challenges arise, leverage their potential as learning opportunities.

IV. AI in Education Resources

General Resources

- [AI Guidance for Schools Toolkit](#) (Teach AI): This toolkit was designed to help education systems create guidance and inform policy. It includes sample school guidance and presentations that can be customized for various groups.
- [AI Guidelines for Youth Mental Health Tools](#) (Children's Hospital Association): This resource includes a recorded on-demand video and materials related to why procurement standards for AI mental health tools are critical to youth safety and well-being, key principals for evaluating vendors to promote privacy, bias mitigation, clinical efficacy and transparency, steps to apply the new guidelines in real-world procurement processes, and practical tools, checklists, and governance compliance and trust. These guidelines were designed to help children's hospitals and school systems make informed, responsible decisions when adopting AI solutions that address youth mental health.
- [AI Literacy Framework](#) (Teach AI): This framework for AI Literacy was open for feedback through August 2025. The Framework includes four domains: Engaging with AI, Creating with AI, Managing AI, and Designing AI and spans K-12
- [Artificial Intelligence and the Future of Teaching and Learning: Insights and Recommendations](#) (U.S. Department of Education, Office of Educational Technology): This report covers a wide range of topics including what AI is, what students should learn about AI, how AI can be used in teaching, and strategies for policy development.
- [Bringing AI to School: Tips for School Leaders](#) (International Society for Technology in Education): This resource for school leaders provides important background about AI, guiding questions, and strategies to successfully bring AI into schools.
- [Empowering Education Leaders: A Toolkit for Safe, Ethical, and Equitable AI Integration](#) (Office of Educational Technology, US Department of Education): This toolkit helps education leaders decide how to use AI in student learning. The AI Report explains key terms, shows major opportunities, and outlines important risks. This document builds on that information and links it to practical guidance for creating school and district AI policies. It also summarizes the Federal laws needed to ensure AI is used safely, securely, and without discrimination.
- [Framework for AI-Powered Learning Environments](#) (National Center for Education and the Economy): This resource includes an overview of trends, underlying assumptions, and guiding principles for integrating AI into education. The resource also includes a framework with short-term (18-20 months) and long-term strategies

(years 2026-30) on integrating AI into education that address students, teachers, schools, districts, and education systems.

- [Review of Guidance from Seven States on AI in Education](#) (Digital Promise): This resource includes a review of AI guidance documents from California, North Carolina, Ohio, Virginia, Washington state, and West Virginia. The review summarizes common themes, identifies opportunities and risks, and offers recommendations about how districts should incorporate AI into education.
- [Shaping the Future of Learning: The Role of AI in Education 4.0](#) (World Economic Forum): This report highlights the potential for AI to create equitable learning opportunities from a global perspective. It includes analysis of the likely impacts of AI on teacher time by role.
- [Teen and Young Adult Perspective on Generative AI](#) (Hopelab, Common Sense Media, The Center for Digital Thriving at Harvard Graduate School of Education): This report shares results from a national survey of digital technology and youth and summarized how youth feel about AI, whether they use it, and what they use it for.

Learning About AI – for Students and Educators

- [AI Basics for K-12 Teachers](#) (Common Sense Education): A foundational course for educators to understand the basics of generative AI and its impact on education. (Requires the creation of a free account.)
- [Artificial Intelligence Lessons](#) (ISTE): Hands-on AI lesson plans for the classroom. Includes lesson plans for all grade levels.
- [AI Literacy Lessons for Grades 6-12](#) (Common Sense Education): Quick lessons for students to think critically about AI and its impact.
- [AI Resources](#) (Code.org): A set of resources including lesson plans, professional development, and videos focused on teaching about AI.
- [Experience AI](#) (Raspberry Pi Foundation): A collection of lesson plans and resources to help introduce AI to students aged 11-14.
- [Teach AI: Classroom Curricula](#) (aiEDU): Resources for educators to teach about AI. Three different sets of activities aimed at different depths of AI learning.

Minnesota Resources

- [Artificial Intelligence](#): Guidance from Minnesota Department of Education (MDE) on issues related to inclusion of artificial intelligence (AI) topics and tools in career and technical education (CTE) courses.
- [Bloomington Public Schools: AI in BPS – Guiding Principles](#); [slide deck](#): This guidance includes information about the ethical use of AI by the district, school, teacher, and student.
- [Minnesota Department of Transportation General Artificial Intelligence Standard](#): Clear guidance to ensure generative AI is used in alignment with the department’s core values, legal requirements, policies, and security practices.
- [Minnesota Generative AI Alliance for Education \(MNGAIA\)](#): This website is the work of a cross-sector group in Minnesota focused on providing resources for AI use in education. The website includes MDE’s Guiding Principles and features resources, tools, a discussion board, a success stories of AI use from across the state.
- [Minnesota IT Services Transparent Artificial Intelligence Governance Alliance](#): Guidance for using AI to improve government services while ensuring ethical, equitable, and secure practices.
- [MSBA/MASA Model Policy 625](#): Model policy to support school districts with developing policy around the responsible use of artificial intelligence.
- [Minnesota State Generative Artificial Intelligence](#): A guidance document on policy intersections, considerations and recommendations.
- [St. Cloud Area School District 742 Guiding Practices for Generative AI](#): These guiding principles discuss the benefits and risks of AI; guidelines for students, teachers, and employees; and academic honesty and AI.

- [Setting AI Expectations: Creating Clarity in Classrooms \(Webinar Recording\)](#): This one hour webinar sponsored by Common Sense Education features two Digital Learning Specialists from Bloomington Public Schools talking about how to co-create expectations with students regarding AI use in classrooms.

Other States

- [Artificial Intelligence Guidance Brief](#) (Kentucky): This resource is a model of guiding principles for Kentucky students and staff in the appropriate and responsible use of AI (specifically generative AI).
- [AI Guidance for West Virginia Schools](#) (West Virginia): This guidance provides support for the use of AI across various roles in West Virginia PK–12 schools, catering to the needs of superintendents, district staff, educators and support staff. The guidance focuses on effectively and safely integrating AI into classroom instruction, school administration and district operations.
- [AI Toolkit](#) (Ohio): This toolkit for Ohio schools includes resources to advance AI literacy and create AI policy, recommendations for teachers to turn policy into practical application, resources for parents, and links to guidelines.
- [Developing Policy and Protocols for the Use of Generative AI in K-12 Classrooms](#) (Oregon): This planning and reflection tool is a resource for school and district leaders to create clear, equitable and meaningful AI policy.
- [Generative AI Implementation Recommendations and Considerations for PK-13 Public Schools](#) (North Carolina): These informational resources are intended to direct responsible implementation of AI tools and guide AI literacy for North Carolina public schools. The guidelines are organized around North Carolina’s Digital Learning Plan.
- [Generative Artificial Intelligence in K-12 Classrooms](#) (Oregon): This resource addresses FAQs related to AI use in classrooms including equity implications, data privacy, and district policy.
- [Human-Centered Artificial Intelligence in Schools](#) (Washington): Washington state created this resource to provide students, educators and school district administrators with the resources and tools they need to understand how to use these technologies effectively, ethically, and safely. It includes AI definitions, sample guidance and policy information (including academic integrity).
- [Leveraging AI in the K-12 Setting](#) (Georgia): This guidance includes general information and best practices for the use of AI in schools. It includes self-check questions for districts that includes considerations for each of the topics in the guidance.
- [State AI Guidance for K-12 Schools](#) (AI for Education): This website features an interactive map that provides links to individual AI guidance for K-12 schools.
- [The AI Roadmap](#) (Maine): This interactive website features an overview of AI, teaching and learning AI modules, a section on AI leadership and how to get additional support.

This document was developed through a collaborative process including expert review and generative AI.